

Saber Ayatollahi, CFA, MSc

Toronto, Canada • +1 (416) 856-1276 • saber.ayatollahi@gmail.com • www.linkedin.com/in/sayatollahi

SUMMARY

Quantitative modeller with ~7 years across ALM, balance-sheet, and derivatives analytics, targeting the Vice President, Capital Markets Treasury Quant mandate at RBC. Led design and implementation of an enterprise multi-asset cash-flow projection and liquidity-gap engine in Python; holds delegated sign-off on valuation, sensitivity, and ALM outputs for institutional portfolios of \$5-25bn per engagement, independently reviewing curve construction, spread calibration, and rates/FX/inflation derivatives pricing. Model governance committee member; daily user of Claude Code and Cursor for AI-assisted model development. CFA, dual MSc (Financial Modelling; Chemical Engineering).

CORE SKILLS

Fixed Income & Derivatives Valuation • Curve Construction & Spread Calibration Review • Interest Rate Risk (IRRBB) & Duration Analytics • Funding, Liquidity & Cash-Flow Projection Modelling • Balance Sheet & ALM Analytics • Model Documentation, Validation & Governance • Stress Testing & Scenario Analysis • Stochastic Simulation & Monte Carlo • Python (pandas, NumPy, SciPy), SQL, VBA, Git version control • AI-Assisted Development (Claude Code, Cursor IDE) • Stakeholder Communication & Model Explanation

EXPERIENCE

Moody's Analytics

Asst. Director - Modelling Services

Toronto, May 2023 - Present

Model Development & Implementation

- Led design and implementation of an enterprise multi-asset cash-flow projection engine supporting base, stress, and reverse-stress scenarios, delivering forward-looking liquidity visibility across ALM and funding horizons.
- Architected configurable time-bucketed liquidity gap analytics (T+1 through multi-year), enabling funding optimization, refinancing risk assessment, and asset-allocation decisions.
- Re-engineered manual spreadsheet workflows into scalable Python analytics pipelines with embedded logging, validation, and auditability controls.
- Built agentic AI development workflows (Claude Code, Cursor IDE) automating code generation, validation scaffolding, and anomaly detection, cutting development cycle time an estimated 30-40% on comparable modules.

Curve, Spread & Derivatives Model Review

- Holds delegated sign-off authority on valuation, sensitivity, and ALM outputs for institutional portfolios of \$5-25bn per engagement (~\$50bn+ cumulative), independently reviewing curve construction, spread calibration, and cross-asset interactions prior to production release.
- Validated derivatives pricing outputs across rates, FX, and inflation, confirming sensitivity consistency at portfolio-level ALM aggregates.
- Oversaw interest rate risk and duration analysis under parallel and non-parallel rate shocks, aligned with IRRBB standards analogous to OSFI B-12 and Basel Committee frameworks.
- Reviewed aggregation logic converting security-level exposures into portfolio-level risk metrics feeding downstream ALM and capital processes.

Model Documentation, Governance & Business Support

- Serves on the model governance committee, contributing to methodology review, model documentation and benchmarking standards, and ongoing model-performance assessment for client-delivered analytics.
- Escalated outputs lacking mathematical or economic defensibility and acted as escalation point for client-critical analytical issues with Heads of Risk and investment stakeholders.
- Prepared analytical summaries for senior stakeholders on interest rate exposure, scenario impacts, and balance sheet sensitivities, translating quantitative output into decision-ready narratives for non-technical audiences.

Client Service Specialist

Toronto, May 2022 - May 2023

- Led onboarding and implementation for U.S. and Canadian pension funds and asset managers migrating to the PFaroe DB (risk & ALM) and PFaroe PM (portfolio management) platforms; validated model outputs post-deployment.
- Translated investment-team requirements into scoped development requests for Product Owners and advised clients on asset, liability, and ALM configurations.

Ernst & Young (EY)

Senior Consultant - IFRS & Financial Risk Transformation

Toronto, Sep 2021 - Apr 2022

- Delivered IFRS 17 and IFRS 9 transformation projects for Canadian and international insurance clients, aligning regulatory requirements with system implementation and financial-reporting frameworks.
- Advised actuarial, finance, IT, and program-management functions on standards adoption and contributed to governance documentation and regulatory-readiness processes.

Ortec Finance Canada

Consultant - Asset-Liability Management & Risk

Toronto, Feb 2019 - Sep 2021

- Built and interpreted stochastic economic scenario generators assessing interest rate risk and funding volatility under base-case and stressed environments.
- Performed cash flow, liquidity, and hedging analysis across interest rate, inflation, and currency risk for international institutional clients.
- Delivered ALM studies analyzing funding ratios, duration mismatch, and long-term balance sheet risk exposures for pension funds and institutional investors.
- Performed asset-only and surplus portfolio optimization on VaR and CVaR using Ortec's GLASS platform, with risk decomposition and near-optimal frontier analysis to test allocation robustness.

EDUCATION & CREDENTIALS

- Chartered Financial Analyst (CFA) - CFA Institute, 2024
- MSc, Financial Modelling - Western University, 2018
- MSc, Chemical Engineering (Full Scholarship) - Western University, 2016